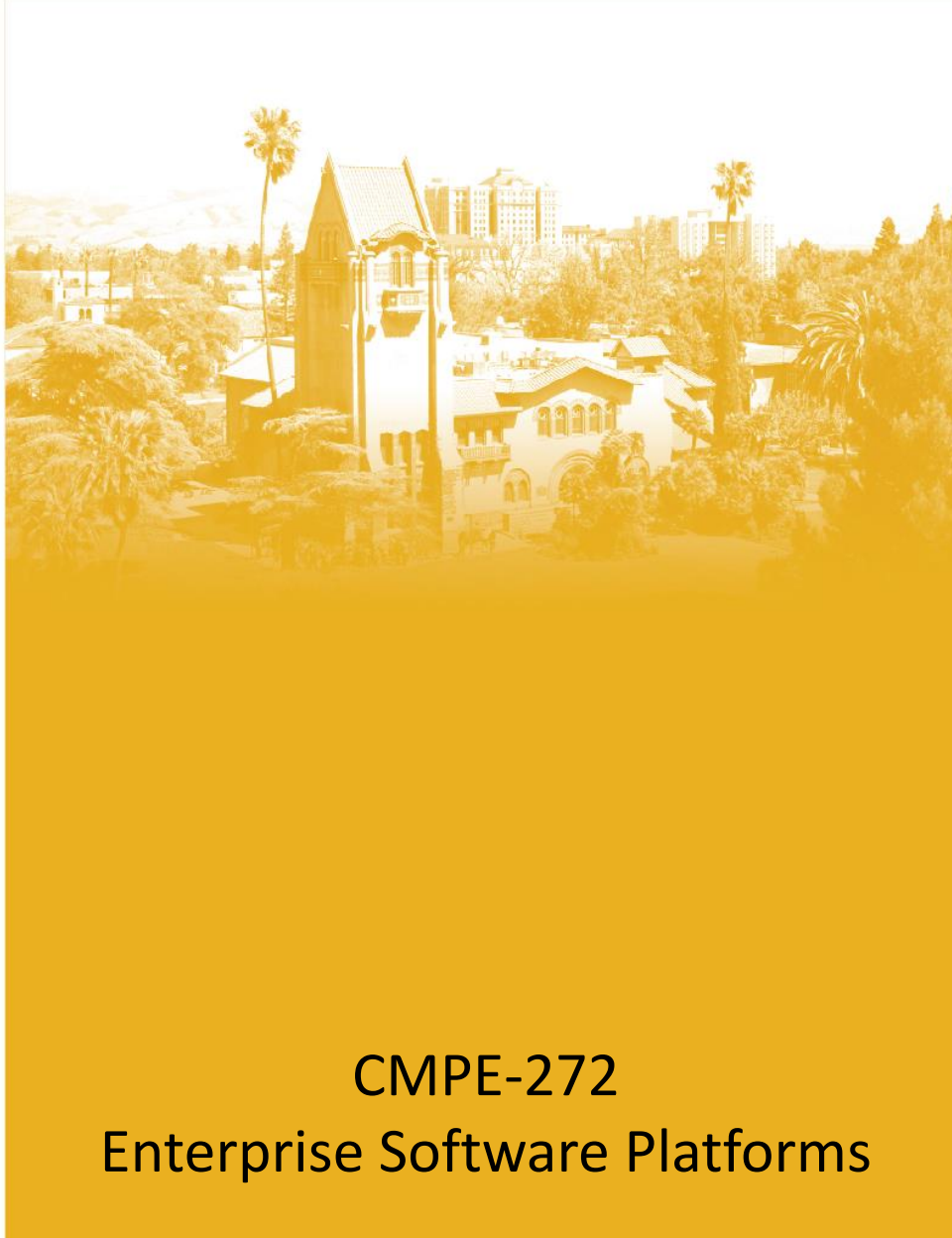


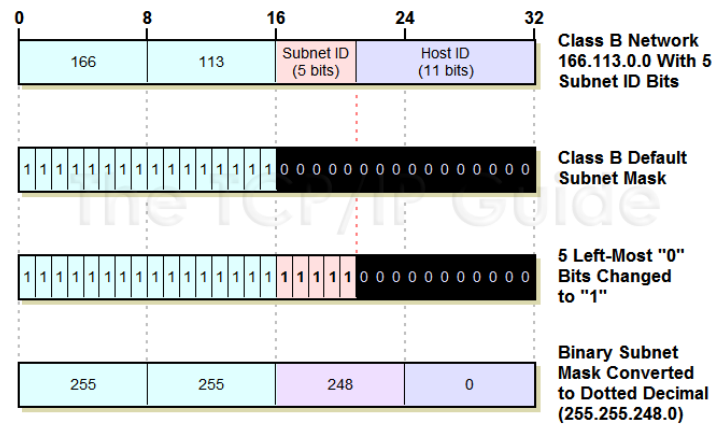


CMPE-272
Enterprise Software Platforms

- OSI Model & IP Data Networks
 - 7 Layers of OSI Model
 - Basic Purpose and functions of various network devices such as routers, switches, bridges and hubs
 - Common services and network functions

- Network Utilities (ping, telnet, ssh, ifconfig, ip – and how to use them)
- IP Addressing (v4)

- Classes, Subnet, Mask, Multicast, VLSM, Notation Reference Sheets:
- [IPv4](#) [IPv6](#)



Basic Networking Concepts

What you should already know

L#	Device Type	OSI Layer	TCP/IP Org.	TCP/IP New	Protocols	PDU
7	Gateway	Application	Application	Application	HTTP, FTP,	Data
6		Presentation			POP,SMTP,	Data
5		Session			DNS, RIP	Data
4		Transport	Transport	Transport	TCP/UDP	Segments
3	Router	Network	Internet	Network	IP, ARP, ICMP, IGMP	Packets
2	Switch/Bridge	Data Link	Link	Data Link	Ethernet,	Frames
1	Hubs/Repeater	Physical		Physical	Token Ring	Bits

Subnets				Decimal to Binary			
CIDR	Subnet Mask	Addresses	Wildcard	Subnet Mask	Wildcard		
/32	255.255.255.255	1	0.0.0.0	255	1111 1111	0 0000 0000	
/31	255.255.255.254	2	0.0.0.1	254	1111 1110	1 0000 0001	
/30	255.255.255.252	4	0.0.0.3	252	1111 1100	3 0000 0011	
/29	255.255.255.248	8	0.0.0.7	248	1111 1000	7 0000 0111	
/28	255.255.255.240	16	0.0.0.15	240	1111 0000	15 0000 1111	
/27	255.255.255.224	32	0.0.0.31	224	1110 0000	31 0001 1111	
/26	255.255.255.192	64	0.0.0.63	192	1100 0000	63 0011 1111	
/25	255.255.255.128	128	0.0.0.127	128	1000 0000	127 0111 1111	
/24	255.255.255.0	256	0.0.0.255	0	0000 0000	255 1111 1111	
/23	255.255.254.0	512	0.0.1.255				
/22	255.255.252.0	1,024	0.0.3.255				
/21	255.255.248.0	2,048	0.0.7.255				
/20	255.255.240.0	4,096	0.0.15.255				
/19	255.255.224.0	8,192	0.0.31.255				
/18	255.255.192.0	16,384	0.0.63.255				
/17	255.255.128.0	32,768	0.0.127.255				
/16	255.255.0.0	65,536	0.0.255.255				
/15	255.254.0.0	131,072	0.1.255.255				
/14	255.252.0.0	262,144	0.3.255.255				
/13	255.248.0.0	524,288	0.7.255.255				
/12	255.240.0.0	1,048,576	0.15.255.255				
/11	255.224.0.0	2,097,152	0.31.255.255				
/10	255.192.0.0	4,194,304	0.63.255.255				
/9	255.128.0.0	8,388,608	0.127.255.255				
/8	255.0.0.0	16,777,216	0.255.255.255				
/7	254.0.0.0	33,554,432	1.255.255.255				
/6	252.0.0.0	67,108,864	3.255.255.255				
/5	248.0.0.0	134,217,728	7.255.255.255				
/4	240.0.0.0	268,435,456	15.255.255.255				
/3	224.0.0.0	536,870,912	31.255.255.255				
/2	192.0.0.0	1,073,741,824	63.255.255.255				
/1	128.0.0.0	2,147,483,648	127.255.255.255				
/0	0.0.0.0	4,294,967,296	255.255.255.255				

Subnet Proportion

Classful Ranges

Class	Range
A	0.0.0.0 - 127.255.255.255
B	128.0.0.0 - 191.255.255.255
C	192.0.0.0 - 223.255.255.255
D	224.0.0.0 - 239.255.255.255
E	240.0.0.0 - 255.255.255.255

Reserved Ranges

Range	Notes
10.0.0.0 - 10.255.255.255	RFC 1918
127.0.0.0 - 127.255.255.255	Localhost
172.16.0.0 - 172.31.255.255	RFC 1918
192.168.0.0 - 192.168.255.255	RFC 1918

QUIC (Quick UDP Internet Connections)



Key Characteristics of QUIC

- QUIC is designed to address several limitations in TCP and provide better support for the modern web, where low-latency connections and secure communication are essential. The major characteristics of QUIC include:
- **UDP-Based Transport**
 - Unlike TCP, which operates at the transport layer of the OSI model and is widely used for reliable transmission of data, QUIC operates over **UDP** (User Datagram Protocol).
 - **UDP** is a lightweight, connectionless protocol. However, UDP lacks features like connection management, reliability, and congestion control. QUIC builds these features on top of UDP, effectively adding reliability, ordering, and encryption to a connectionless transport layer.
 - The use of UDP allows QUIC to bypass many of the limitations of TCP, especially those related to latency and slow connection setup, while preserving the advantages of reliability.

Multiplexing Without Head-of-Line Blocking

- One of the significant problems with HTTP/2 over TCP is head-of-line blocking. If one packet is lost in TCP, all subsequent packets must wait for that lost packet to be retransmitted, causing delays across all streams using the same TCP connection.
- QUIC solves this by multiplexing multiple streams over a single connection and handling packet loss on a per-stream basis. This means that if a packet is lost for one stream, other streams can continue without delay, significantly improving performance in environments with packet loss.

What is Switching?

- A “switch” is a hard-wired connection between 2 wires
- A switch in a network device is a nailed-down connection between input and output
- A packet is switched if the forwarding decision is pre-determined, i.e. trivial per-packet processing necessary
- Switching in packet world implies label switching
- Label switching (ATM) was an early h/w forwarding implementation
- Switching therefore came to mean forwarding in h/w, as opposed to s/w
- Switching has morphed into a marketing term

What is a VLAN?

- **Virtual LAN (VLAN)** – a group of interfaces on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments:
- VLANs are logical partitions of flooding domains
- Many different ways to partition VLANs.

Some examples are:

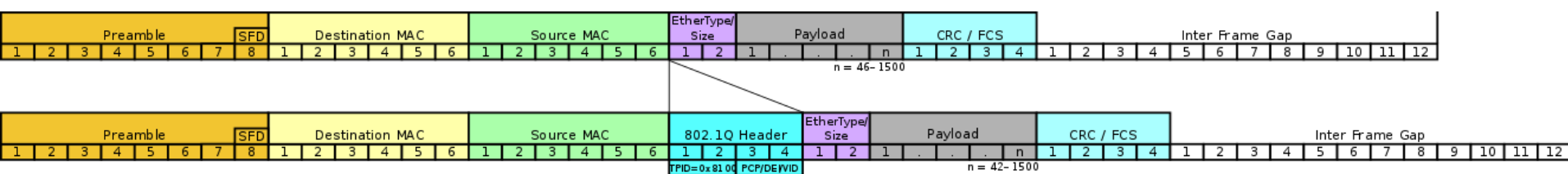
- MAC address
- Port
- Protocol

VLAN Trunk

- A port which carries the traffic of multiple VLANs through the use of encapsulation or other explicit technique.
Examples are ISL, IEEE802.1q, LANE
- Normally used as a link between forwarding devices such as switches and routers
- Each frame transmitted on a trunk link is "tagged" as belonging to one and only one VLAN

IEEE 802.1q

- AKA, “dot1q”
- 4 byte “Tag” inserted after the regular Ethernet SA and before EtherType/Length field
- First 2 bytes shows a value of 0x8100; serves as a 802.1q EtherType
- Next 16 bits encapsulate:
 - 12-bit VLAN ID
 - 3 bits of priority
 - 1bit of Canonical Format Identifier (CFI) - compatibility between Token Ring type and Ethernet type networks



IEEE 802.1q cont'd

- Both tagged and untagged frames are allowed on dot1q
- Untagged frames transmitted and received on dot1q are associated with the Native VLAN
- Null VLAN id with priority tagged frames are considered as untagged frames

Layer2 Forwarding

- Unlike IGPs (Interior Gateway Layer3 Routing Protocols), Layer2 switching does not have a protocol to setup forwarding tables
- Layer2 Forwarding table (CAM table/MAC table) is build by watching source MAC address in data packets
- Layer2 Forwarding happens based on destination MAC, if entry not present in CAM then **Unknown Unicast flooding** is used

Spanning Tree

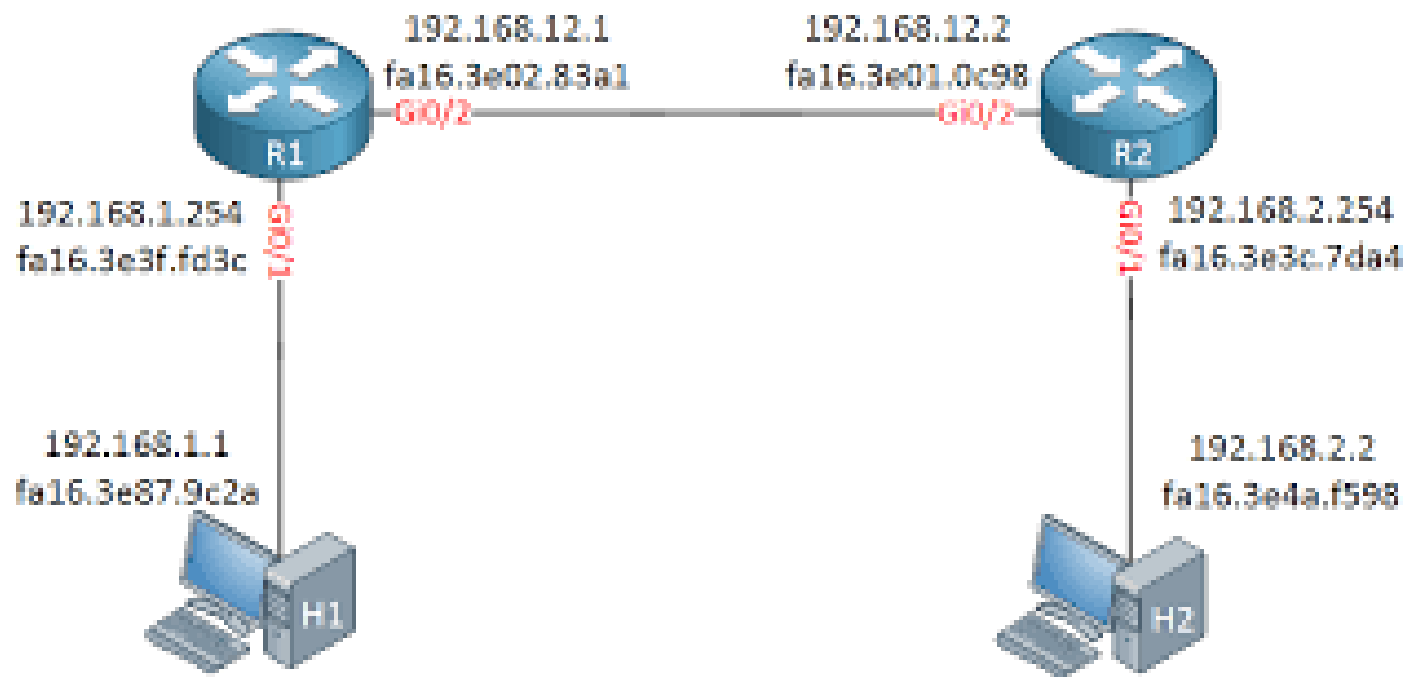
- The **Spanning Tree Protocol (STP)** is a [network protocol](#) that builds a logical loop-free [topology](#) for [Ethernet networks](#). The basic function of STP is to prevent [bridge loops](#) and the [broadcast radiation](#) that results from them. Spanning tree also allows a [network design](#) to include backup links to provide [fault tolerance](#) if an active link fails.
- As the name suggests, STP creates a [spanning tree](#) within a network of connected layer-2 [bridges](#), and disables those links that are not part of the spanning tree, leaving a single active path between any two network nodes. STP is based on an [algorithm](#) that was invented by [Radia Perlman](#) while she was working for [Digital Equipment Corporation](#).^{[1][2]}
- In 2001, the [IEEE](#) introduced **Rapid Spanning Tree Protocol (RSTP)** as 802.1w. RSTP provides significantly faster spanning tree convergence after a topology change, introducing new convergence behaviors and bridge port roles to do this. RSTP was designed to be backwards-compatible with standard STP.

Algoryhme by Radia Perlman

I think that I shall never see
A graph more lovely than a tree.
A tree whose crucial property
Is loop-free connectivity.
A tree that must be sure to span
So packets can reach every LAN.
First, the root must be selected.
By ID, it is elected.
Least-cost paths from root are traced.
In the tree, these paths are placed.
A mesh is made by folks like me,
Then bridges find a spanning tree.



[Just don't call her
the mother of the
internet](#)



LAYER 3: IP ROUTING

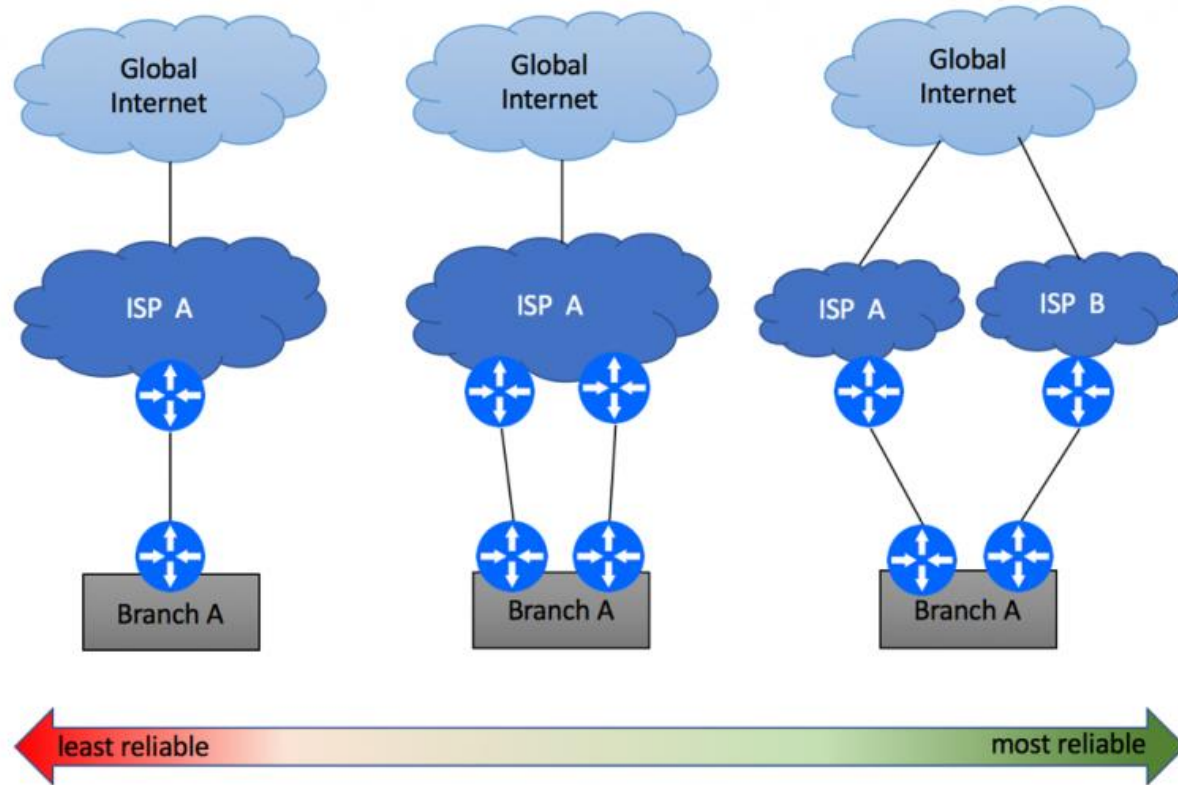
IGP vs. EGP

- An AS (Autonomous System) is a single SP's network
- The IGP protocols in use today are: RIP/**RIPNG**, IGRP, **EIGRP**, **IS-IS**, **OSPF**
- There is only 1 EGP: **BGP**
- IGP's route within an AS or enterprise
- EGP's route between AS's

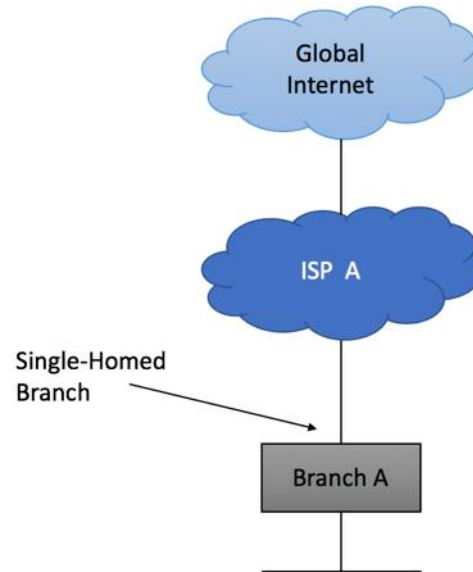
Internet inter-AS routing: BGP

- **BGP (Border Gateway Protocol):** *the* de facto inter-domain routing protocol
 - “glue that holds the Internet together”
- allows subnet to advertise its existence, and the destinations it can reach, to rest of Internet: *“I am here, here is who I can reach, and how”*
- BGP provides each AS a means to:
 - **eBGP:** obtain subnet reachability information from neighboring ASes
 - **iBGP:** propagate reachability information to all AS-internal routers.
 - determine “good” routes to other networks based on reachability information and *policy*

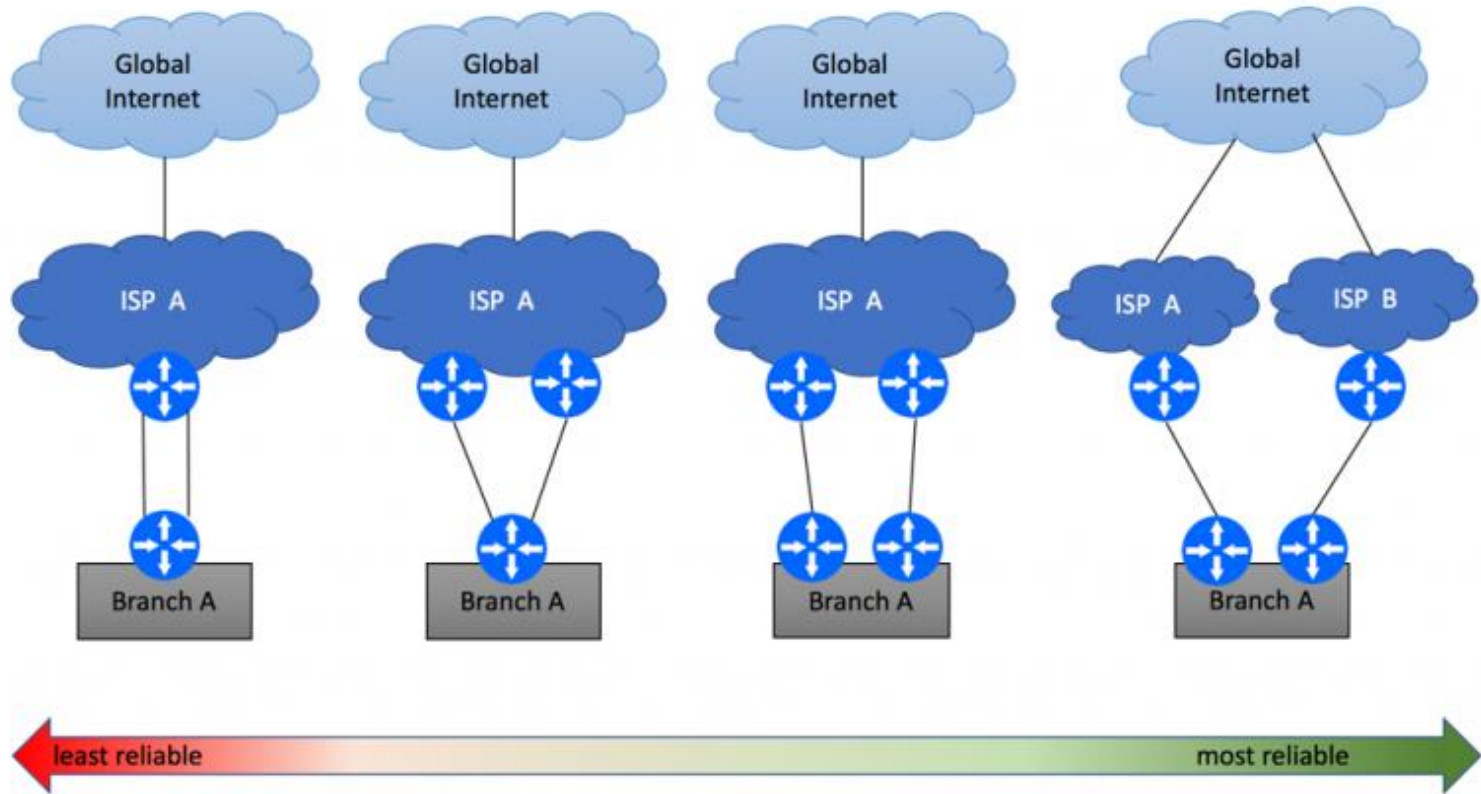
BGP For Enterprise Networks



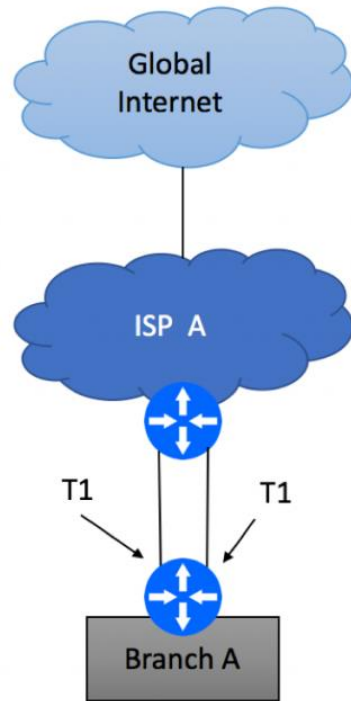
Single-homed network



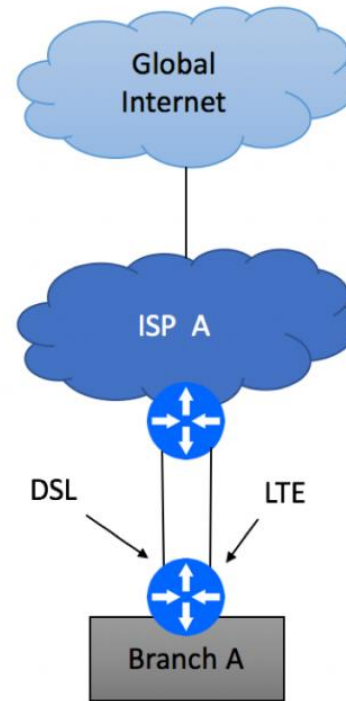
Multi-Homing



Scenario 1 – Same PE / CE

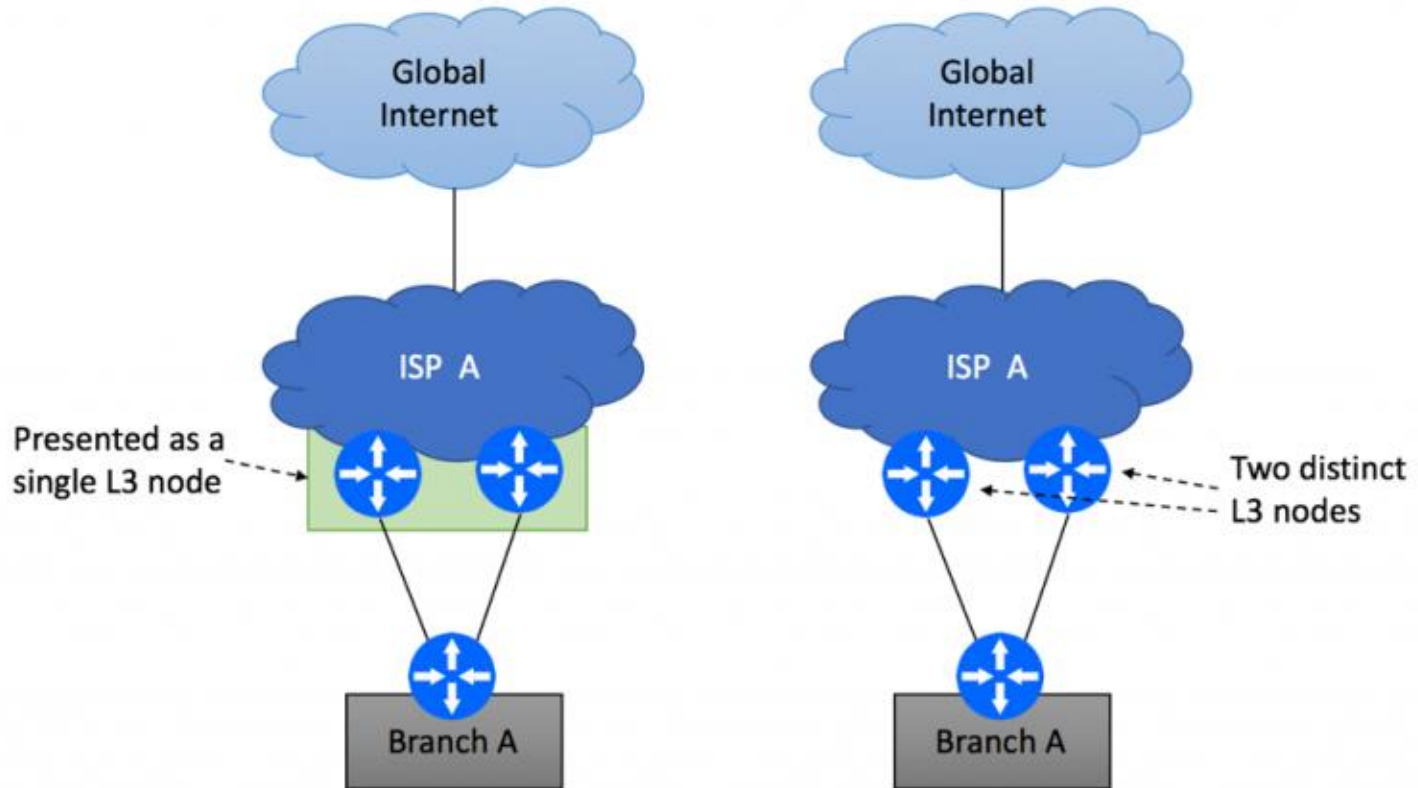


HA using common
access technology

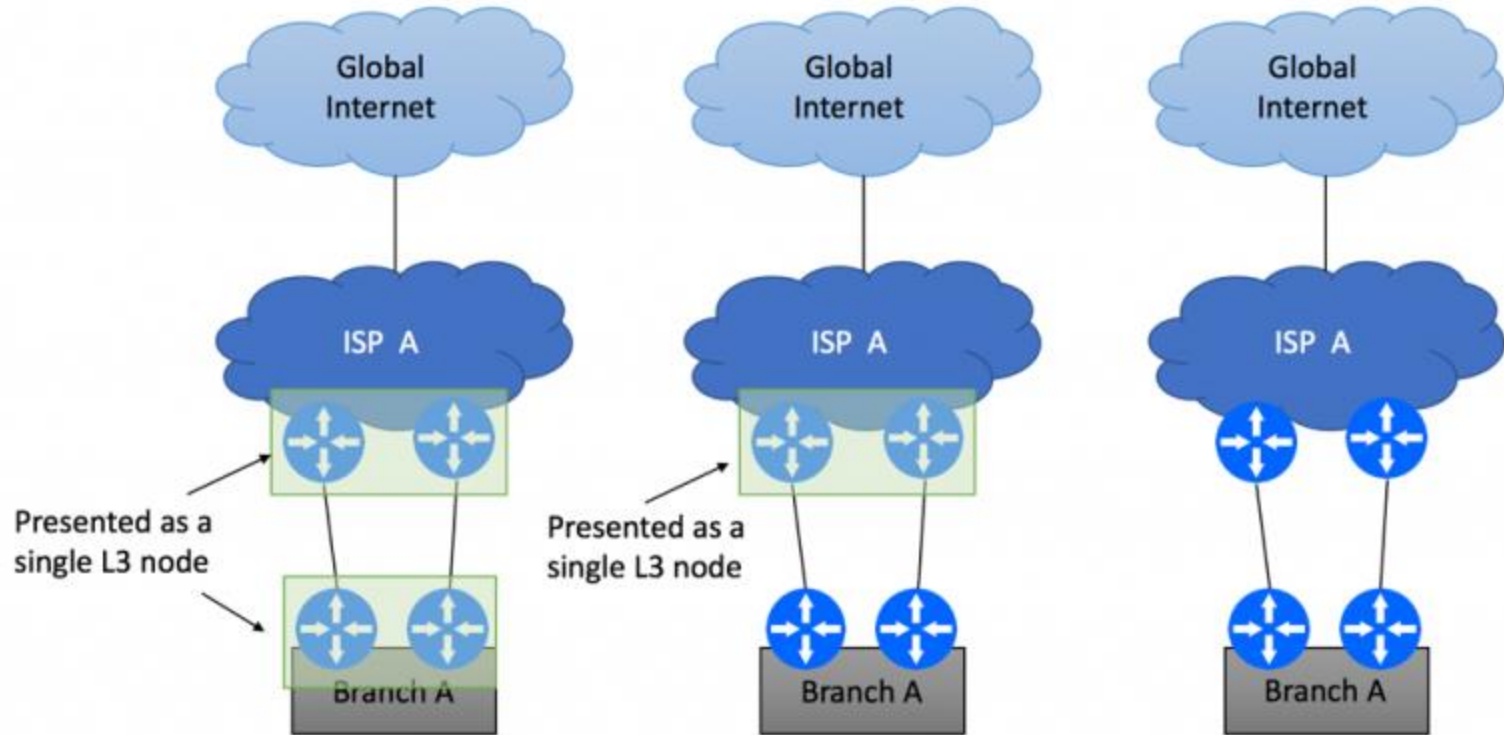


HA using different
access technologies

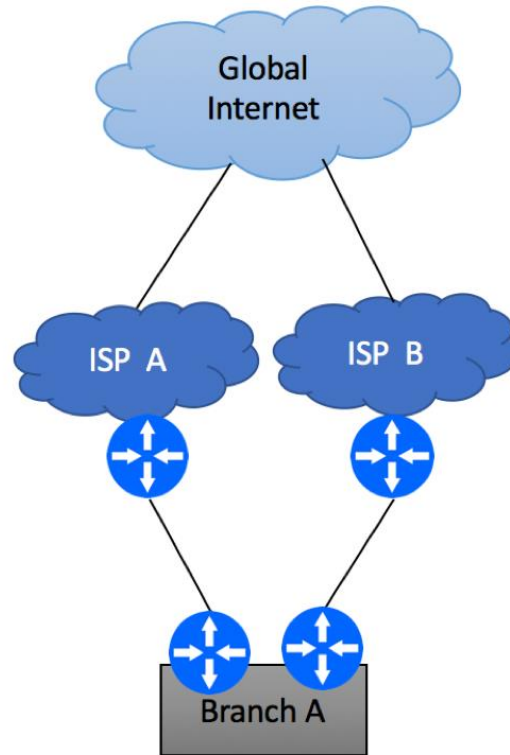
Scenario 2 – Different PEs / Single CE



Scenario 3 – Different PEs and CEs



Scenario 4 – Multiple ISPs

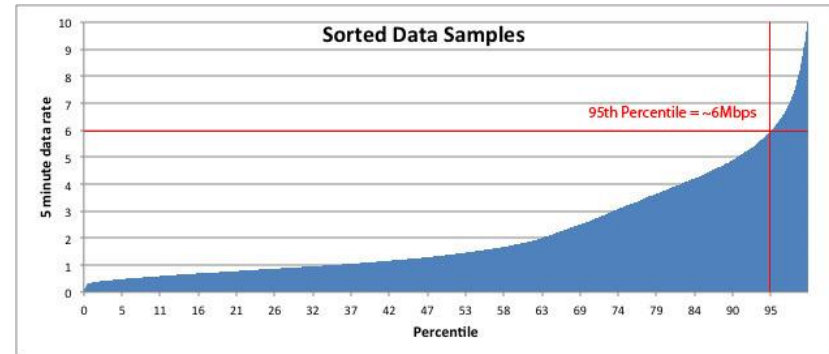
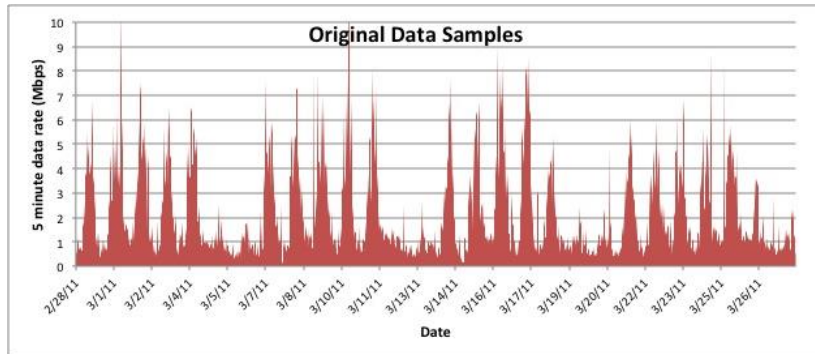


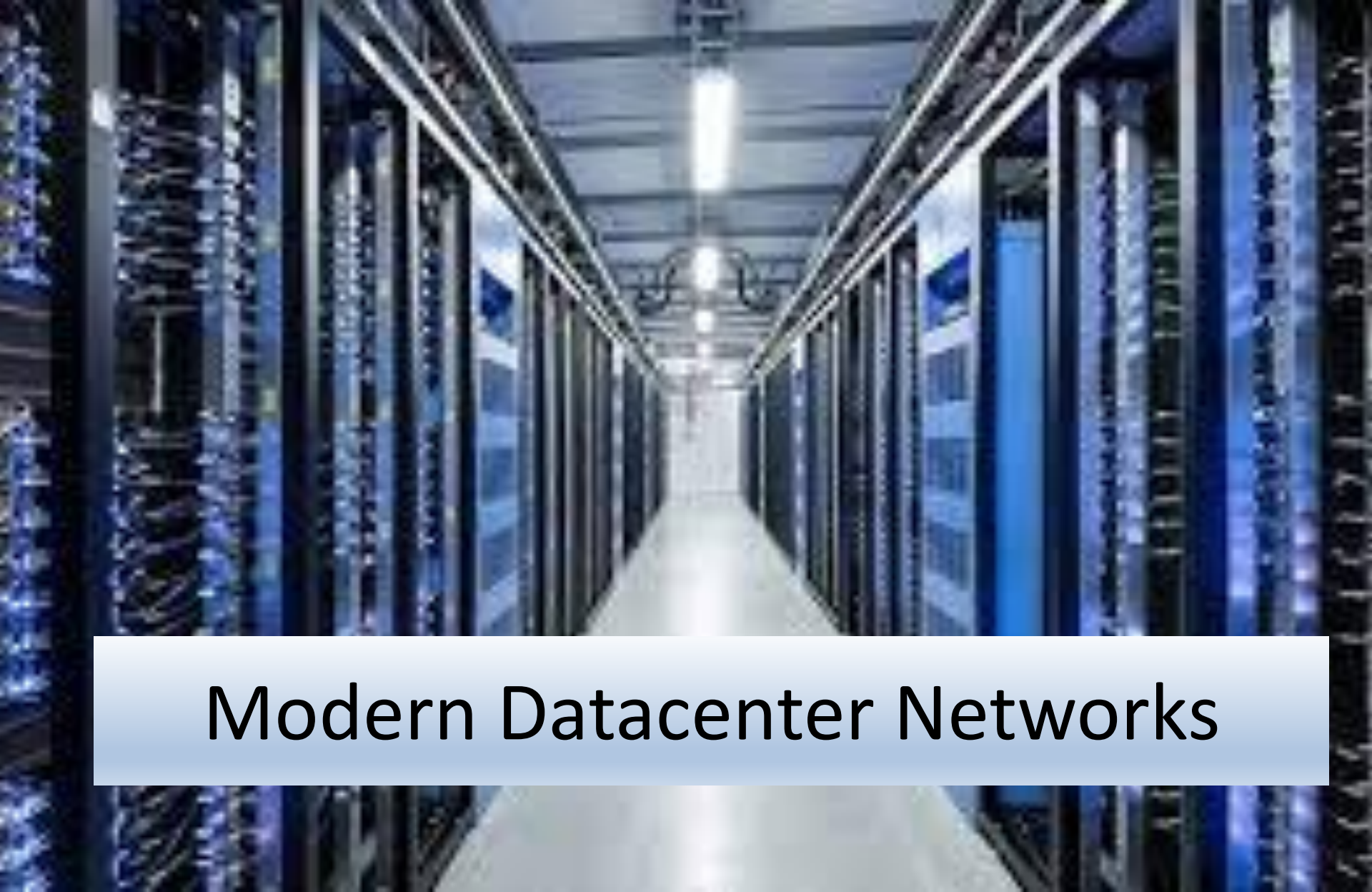
Datacenter Billing

- **Committed Information Rate (CIR)**
 - CIR is a guarantee that the port will always have the bandwidth you're paying for available to it
 - Traditional TDM circuits such as Frame Relay use this method for provisioning so a customer always knows they are going to get
 - Has a hard cap without any allowance for bursting
 - This allows the ISP to tightly control the amount of bandwidth entering and leaving their network, which is required on large broadband networks where a given segment is likely oversubscribed rather than overprovisioned
 - Often Asymmetric B/W (Up/Down)

P95: 95th Percentile Metering

- **P95** is a bandwidth usage metering scheme that allows a customer to burst beyond their committed base rate, and still provides the carrier with the ability to scale their billing with the cost of the infrastructure and transit commits (if any)
- Carriers sample the amount of data transferred on a customer's port(s) every 5 minutes and use that value to derive a data rate (typically in megabits per second or Mbps) for that 5 minute interval





Modern Datacenter Networks

Management, Control Plane & Data Plane

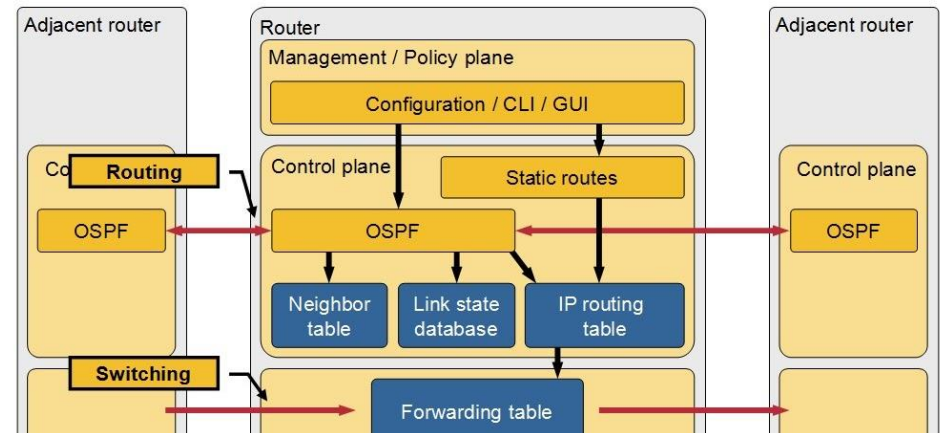
Control Plane

- Makes decisions about where traffic is sent
- Control plane packets are *destined to* or locally *originated by* the router itself
- The control plane functions include the system configuration, management, and exchange of routing table information
- The route controller exchanges the topology information with other routers and constructs a routing table based on a routing protocol, for example, RIP, OSPF or BGP
- Control plane packets are processed by the router to update the routing table information.
- It is the **Signalling** of the network
- Since the control functions are not performed on each arriving individual packet, they do not have a strict speed constraint and are less time-critical

Data Plane

- Also known as **Forwarding Plane**
- Forwards traffic to the next hop along the path to the selected destination network according to control plane logic
- Data plane packets go **through** the router
- The routers/switches use what the control plane built to dispose of incoming and outgoing frames and packets

Management, Control and Data Planes



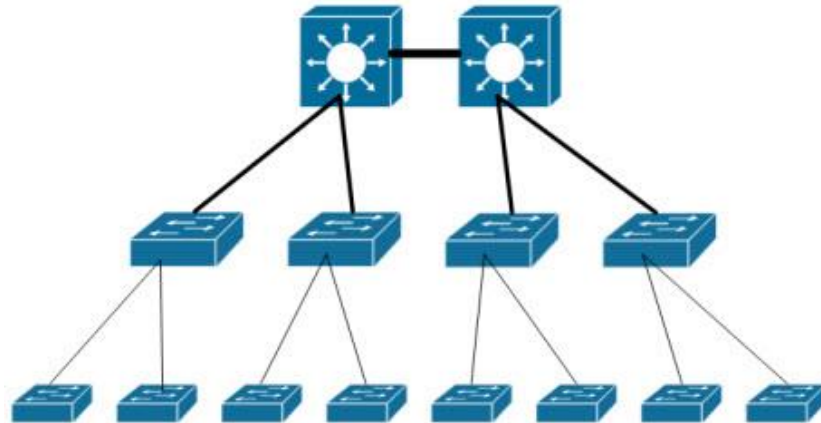
Clos networks

- First researched in the mid-1950s as a method to switch telephone calls. Clos networks evolved into **crossbar topologies** and eventually into chassis-based Ethernet switches using a crossbar switching fabric. Now Clos networks are being used in modern data center networking architectures to achieve high performance and resiliency
- Charles Clos was a researcher at Bell Laboratories in the 1950s. He published a paper titled "A Study of Non-blocking Switching Networks" in the Bell System Technical Journal in 1953
- Clos networks evolved into crossbar topologies and eventually into chassis-based Ethernet switches using a crossbar switching fabric
- Now Clos networks are being used in modern data center networking architectures to achieve high performance and resiliency
- Crossbar fabrics fell out of favor because they were subject to [Head Of Line](#) (HOL) blocking due to input queue limitations. Over time, Ethernet switches were developed that had input and output queues on all the interfaces
- Modern Ethernet switches have more advanced fabric technologies, output queuing and priority-based flow control so they can now achieve non-blocking performance



Traditional Data Center Architecture

- Modern Ethernet switches have more advanced fabric technologies, output queuing and priority-based flow control so they can now achieve non-blocking performance. With these technical enhancements, switches can now support guaranteed bandwidth connectivity for protocols like [Fiber Channel over Ethernet](#) (FCoE) using 10 Gigabit Ethernet linksues on all the interfaces
- the "[fat tree](#)" model



The problem with traditional networks built using the [spanning-tree protocol](#) or layer-3 routed core networks is that a single "best path" is chosen from a set of alternative paths. All data traffic takes that "best path" until the point that it gets congested then packets are dropped. The alternative paths are not utilized because they topology algorithm deemed them to be less desirable or removed to prevent loops from forming.

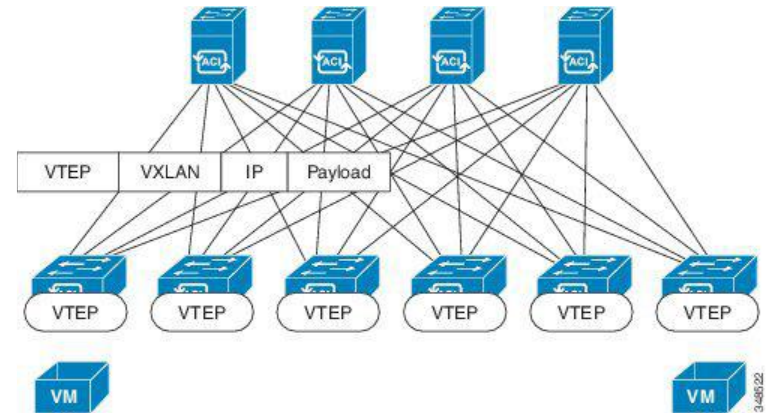
Leaf/Spine Clos Architecture

- Clos networks have now made their second reappearance in modern data center switching topologies
- Rather than being a fabric within a single device, the Clos network now manifests itself in the way that the switches are interconnected
- Data center networks now comprised of top of rack (ToR) switches are the leaf switches attached to the core switches which represent the spine
- Leaf switches are not connected to each other and spine switches only connect to the leaf switches (or an upstream core device)
- In this Spine-Leaf architecture, the number of uplinks from the leaf switch equals the number of spine switches. Similarly, the number of downlinks from the spine equal the number of leaf switches.



Example: Cisco ACI

- ACI fabric decouples the tenant endpoint address, its identifier, from the location of the endpoint that is defined by its locator or VXLAN tunnel endpoint (VTEP) address
- Forwarding within the fabric is between VTEPs
- Mapping of the internal tenant MAC or IP address to a location is performed by the VTEP using a distributed mapping database
- With this model, we can have a full mesh, loop-free topology without the need to use the spanning-tree protocol to prevent loops



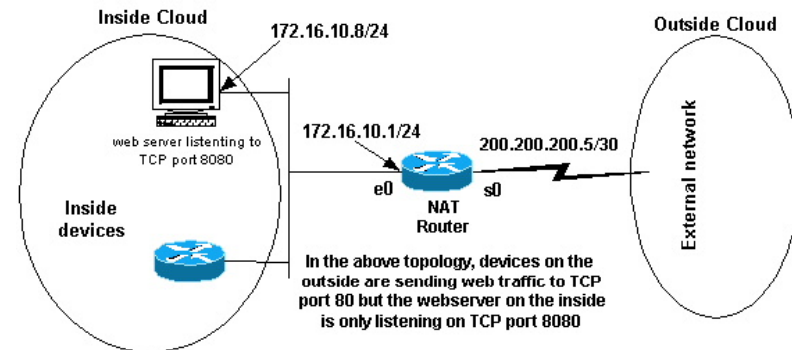
Network Access Translation (NAT)

NAT Concepts

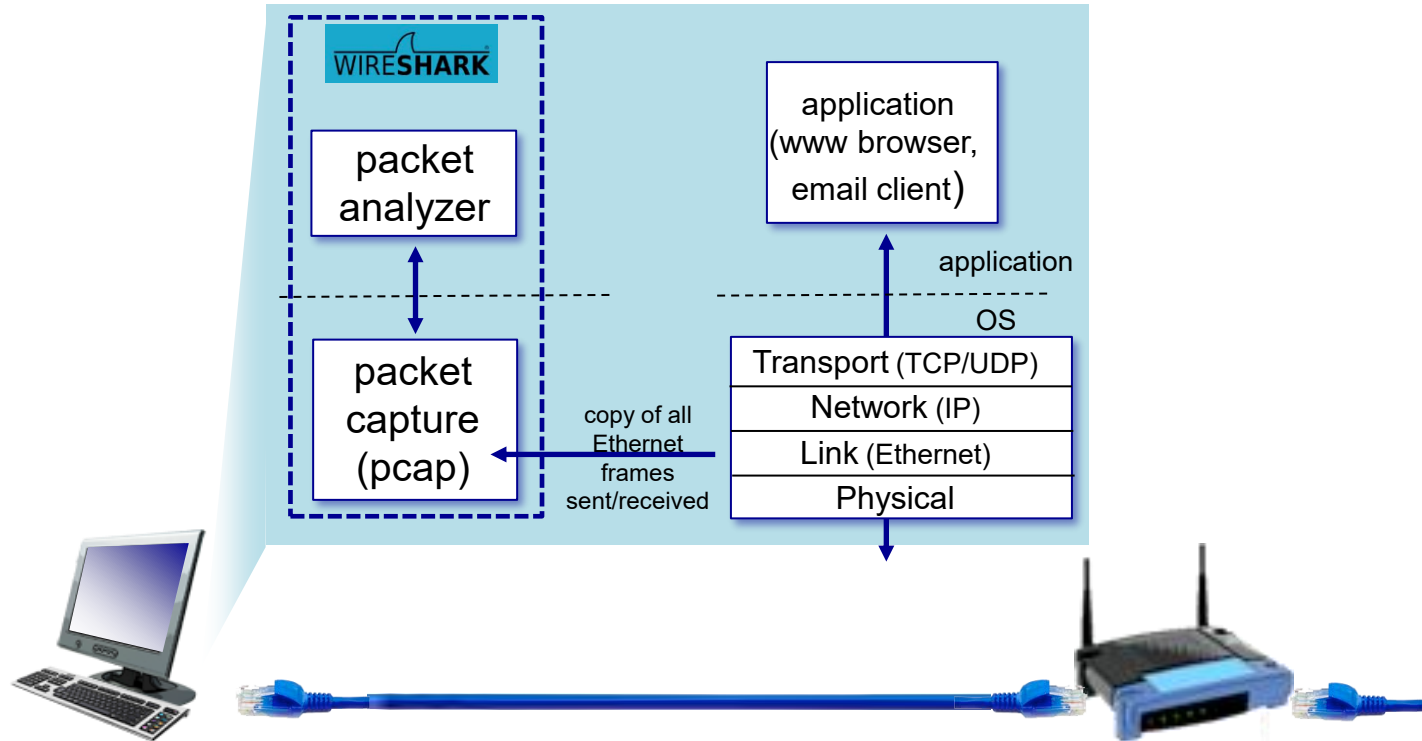
- **Inside and Outside Addresses**
 - **inside local address:** This is the inside address as it is seen and used within the organizational network.
 - **inside global address:** This is the inside address as it is seen and used on the outside of the organizational network.
 - **outside local address:** This is the outside address as it is seen and used within the organizational network.
 - **outside global address:** This is the outside address as it is seen and used on the outside of the organizational network.

- **NAT Types**

- **Static address translation (Static NAT):** This type of NAT is used when a single inside address needs to be translated to a single outside address or vice versa.
- **Dynamic address translation (Dynamic NAT):** This type of NAT is used when an inside address (or addresses) need to be translated to an outside pool of addresses or vice versa.
- **Overloading (Port Address Translation (PAT)):** This type of NAT is a variation on dynamic NAT. With dynamic NAT, there is always a one to one relationship between inside and outside addresses; if the outside address pool is ever exhausted, traffic from the next addresses requesting translation will be dropped. With overloading, instead of a one to one relationship, traffic is translated and given a specific outside port number to communicate with; in this situation, many internal hosts can be using the same outside address while utilizing different port numbers.



Wireshark



Readings and Project

- I. Review/Study this slide deck and links/references
- II. Read: Fowler (ch. 2), Newman (ch. 3), and:
<http://martinfowler.com/articles/microservices.html>